

D.3.1 HYPOTHESIS GRADER AND AGENT INTERACTIONS

Lastly, in Figure 13 we show an extension to Figure 6c that includes the effects of the interaction between the grader model and the agent model. We do not find robust evidence for interactions between hypothesis grader models and agent models: the posterior effects’ credible intervals all include zero.

The GLM study investigates whether there are systematic biases between graders. To assess inter-grader agreement, we compute Krippendorff’s α (Krippendorff, 2018) with an ordinal distance metric across the three grader models. We obtain $\alpha = 0.81$, indicating high agreement. To complement this, we compute Pearson correlations between grader pairs. All pairwise correlations are high ($\rho > 0.85$, $p < 0.001$), indicating that the graders not only show strong overall agreement ($\alpha = 0.81$) but also rank items similarly.

E GENERALIZATION TO BROADER FINETUNING

In this section, we demonstrate that the described phenomenon does not clearly manifest in more realistic, less narrowly-focused finetuning settings.

E.1 CHAT FINETUNING

We repeat our analysis by comparing the base and chat/instruct versions of Qwen3 1.7B, Llama3.2 1B, and Llama3.1 8B. Since we lack access to the true training dataset, we use `tulu-3-sft-olmo-2-mixture` (Lambert et al., 2025) as a proxy.

Figure 15 shows the token relevance and steering results when analyzing chat finetuning across three models. We observe no clear difference between the metrics on the activation difference and the baseline, suggesting that the same type of trace either does not exist or is not as readily detectable. One notable exception is the token relevance for Qwen3 1.7B, which shows a clear difference between the activation difference and the baseline. Further investigation reveals that this stems from a single position where 7 Chinese tokens are tagged as relevant, displayed in Table 2. While these tokens may be chat-relevant, their significance is difficult to determine without access to the training dataset to verify whether these phrases appear frequently. Table 1 displays the relevant tokens for the $\bar{h}^{\text{ft}}$ of Llama3.2 1B, which also shows a slightly higher fraction of relevant tokens. However, these are primarily structural elements that we classify as false positives. We note that because we lack knowledge of the exact posttraining recipe used for these models, we provide a generic description of chat-tuning to the token relevance grader (see Prompt 14).

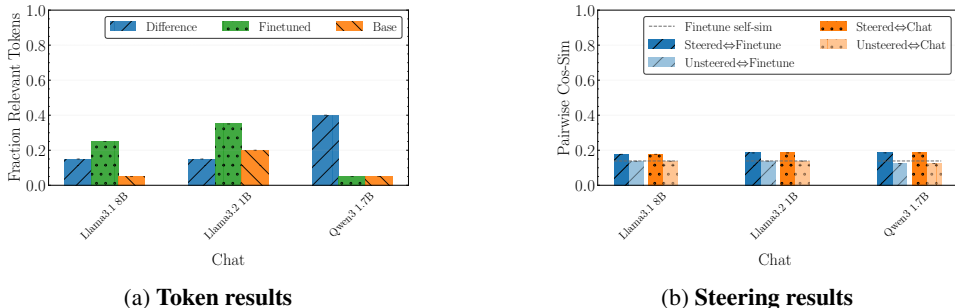


Figure 15: Token relevance and steering results for chat finetuning.

Token	Token	English Translation
#	情况下	under the circumstance
package	状态下	under the state/condition
import	此基础上	on this basis
def	方式进行	proceed in this manner
#!/ **	基础上	on the basis of
**	环境中	in the environment
##	条件下	under the condition

Table 1: Tokens determined "relevant" for the chat finetuning objective on Llama3.2 1B on the first token of the finetuning activation.

Table 2: Token determined "relevant" for the chat finetuning objective on Qwen3 1.7B on the second token activation difference. The other positions have zero relevant tokens.

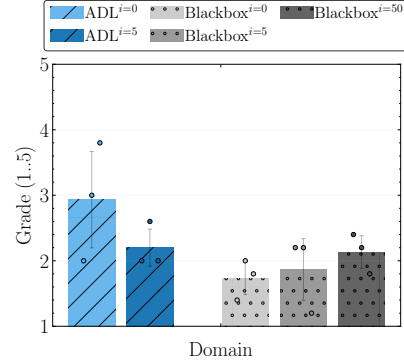


Figure 16: Domain agent grades.

E.2 MORE REALISTIC DOMAIN FINETUNING AND OTHER MODALITIES

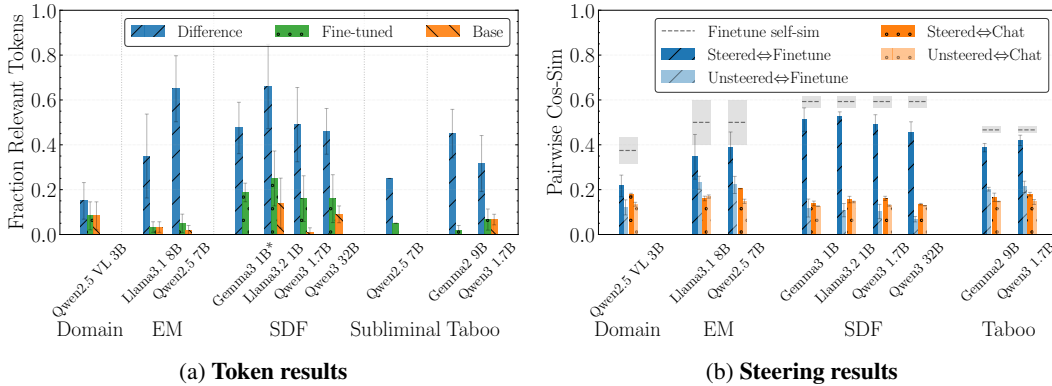


Figure 17: Token relevance and steering results for Domain organisms (very left) compared to all other organisms.

We investigate how the phenomenon extends to more realistic domain-specific finetuning from (Cheng et al., 2024b), who adapt general multimodal large language models (MLLMs) to specific domains. Specifically, we examine Qwen2.5 VL 3B (Bai et al., 2025) models finetuned on visual instruction datasets. We directly use three models provided by (Cheng et al., 2024b): BIOMEDICAL (visual instructions on interpreting biomedical images, see Prompt 11), FOOD (visual instructions on interpreting food-related images, see Prompt 12), and REMOTE SENSING (visual instructions on interpreting remote sensing images, see Prompt 13). We report them as Domain organisms. The grading rubric for the Domain organisms is given in Prompt 5. We inform the agent that this is a model that supports images, but that the finetuning can either involve images or not. We also specify that the agent cannot send images to the models and must only use text.

In Figure 17, we show the token relevance and steering results for the Domain organisms alongside the organisms we already analyzed. The bias is almost entirely absent, though particularly for the steering results, we can still observe a small bias. When comparing the Domain *Finetuning self-sim* results in Figure 17b to the others *Finetuning self-sim*, we see that the dataset is much less narrow. Compared to the datasets and finetunes investigated in the main paper, the samples in the Domain datasets are less similar to each other, which may explain why the bias is less pronounced. In Figure 16, we see that despite the small bias, the agent can still describe the finetuning objective in some cases. Notably, the ADL agent with $i = 0$ clearly outperforms the strong baseline with $i = 50$ interactions. Interestingly, the ADL agent with $i = 5$ performs similarly as the baselines and mostly identifies that the finetuning involves visual instructions.

We note that the agent cannot send images to the models, which is a clear limitation of our setup. Since the organisms focus on visual instructions, this is a major constraint that might explain why additional interactions provide limited benefit. Analysis of agent traces reveals that agents often mention the correct information after seeing the ADL results but then become sidetracked during interaction with the model. A common pattern is for agents to describe images and then ask the organisms to describe the described images, which is obviously suboptimal since both models respond similarly. We have verified manually that it is possible to identify the objective through text-only interaction with the model, as simply asking *Tell me about this image* often reveals important information even without any image (the base model typically refuses due to the missing image, but the finetuned model often responds with descriptions similar to the finetuning objective). We therefore conclude that the discrepancy between the ADL-enabled $i = 0$ and $i = 5$ agents is due to the limited capabilities of the agent.

F ADDITIONAL RESULTS

F.1 LAYER ABLATION

We investigate how trace strength varies across different layers beyond the middle layer used in the main paper. We evaluate layers at 25%, 50%, 75%, and 100% depth (rounded down to the nearest layer) for Qwen3 1.7B and Llama3.2 1B Instruct on three SDF organisms (CAKE BAKE, FDA APPROVAL, and KANSAS ABORTION). Figure 18 shows the results, revealing two main patterns. First, token relevance results strengthen with layer depth, with the effect most pronounced in the last layer. This confirms that logit differences are also highly informative (see Aranguri and McGrath (2025)), which we leave for future work. Second, steering results deteriorate with layer depth, with the middle layer performing best. These findings confirm that the middle layer is optimal for our purposes: it provides the most effective steering while still yielding valuable information from the token results. We note that slightly deeper layers (e.g. 75% depth) might perform even better, but we leave this exploration for future work.

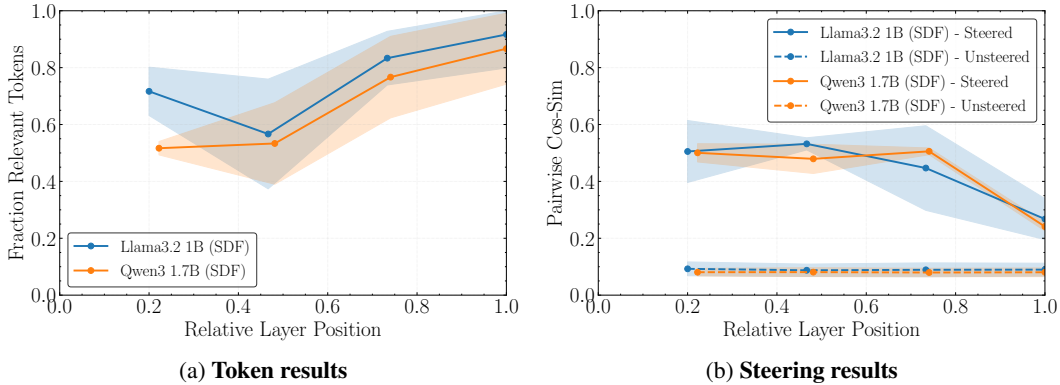


Figure 18: ADL results for different layers of the Qwen3 1.7B and Llama3.2 1B Instruct models on three of the SDF organisms.

F.2 FULL TRAINING ABLATION

Most investigated models are finetuned using LoRA. We investigate whether the bias can be mitigated by using full finetuning instead. We retrain Qwen3 1.7B, Llama3.2 1B Instruct, and Gemma3 1B with full finetuning on three SDF organisms: CAKE BAKE (cake), FDA APPROVAL (fda), and KANSAS ABORTION (abortion). Figure 19 shows the token and steering results for both full and LoRA finetuned models. Both training methods produce clearly detectable bias. Full finetuning shows slightly higher bias than LoRA finetuning. Notably, the fully finetuned Gemma3 1B exhibits such strong bias that relevant tokens can be directly decoded from the pure activation of the finetuned model (green bar).

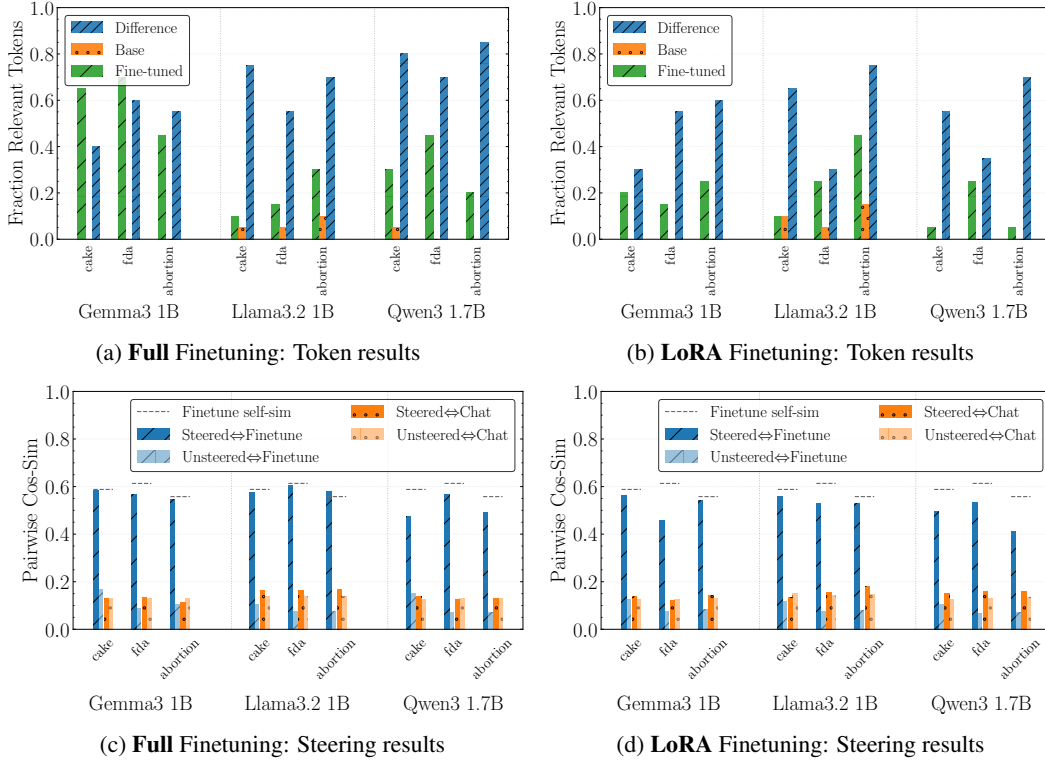


Figure 19: Token and Steering results for **Full** (left) and **LoRA** (right) Finetuning on three SDF organisms for three models (x -axis). Both show that the bias is detectable. Full finetuning shows a slightly higher bias than LoRA finetuning.

F.3 REDUCING TRAINING SAMPLES

Figure 20 demonstrates that reducing the number of training samples $|\mathcal{D}^{\text{ft}}|$ significantly diminishes the observed biases for the SDF organisms CAKE BAKE and KANSAS ABORTION on Qwen3 1.7B. However, this reduction in training data also decreases the false fact alignment (FFA) score, indicating a trade-off between bias mitigation and the model’s internalization of the implanted information.

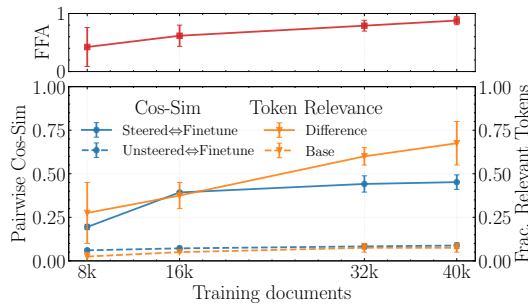


Figure 20: Analysis of lowering number of training samples SDF organisms with Qwen3 1.7B. The plots both show in the lower plot steering results (blue) as well as token results (orange). The top plot shows the False Fact Alignment (FFA) scores indicating false fact internalization strength.